

Using QEMU

for Embedded Systems Development

BUILDING QEMU
FROM SOURCE
CREATING A NEW
VIRTUAL MACHINE
QEMU NETWORKING



Part—1

In this series of articles, we will explore the basics of QEMU, OS installation, QEMU networking and embedded system development for the ARM architecture. In this first part, let's begin with the basics.

When I first began using computers, I was amazed. It required just a mouse-click to play songs, movies or games to entertain every age. It was like magic to me! Over time, I learnt about compiled programs, and source code. My curiosity very soon made source code my passion. Even though compiled software packages are easily available, I love compiling from source. And that is just what I do for QEMU.

QEMU is one of the best emulators out there. Still, very few people use its full capabilities. Though we deal with the basics in this article, look forward to some interesting stuff later in the series!

Building QEMU from source

The first thing is to download the QEMU source code; the current distribution is at <http://download.savannah.gnu.org/releases/qemu/qemu-0.14.0.tar.gz>. Extract the tar ball and go to the extracted directory:

```
$ tar -zxvf qemu-0.14.0.tar.gz
$ cd qemu-0.14.0
```

Run the configuration script. We will build QEMU for i386. (It can be built for other architectures too, like ARM, PPC, SPARC, etc.) Let's install the Ubuntu distro in the virtual machine—that's the reason we've chosen to build QEMU for the i386 architecture:

```
$ ./configure --target-list=i386-softmmu
```

Hopefully, you will not run into any trouble during the *configure* script run. If there's any issue, it will probably be some missing library or header files, which you can look for, and install. Once you are done with the *configure* script, compile the source code with the *make* command.

After compilation, QEMU binaries should be installed in their proper locations. On my Fedora system, I used the *su* command to get the necessary root privileges and install the binaries:

```
$ su
Password:
# make install
```

To confirm that QEMU has been successfully installed, run *qemu*, and a pop-up window like what appears in Figure 1 will confirm the successful installation of QEMU.

Creating a new virtual machine

If you are familiar with other virtualisation software, you might wonder how to go about creating a new VM. The first step is to create the hard disk image for the VM. So let's install the Ubuntu 9.10 OS in the VM; a disk image of 10 GB is sufficient for this purpose. To create it, run the following code:

```
$ qemu-img create ubuntu.img 10G
$ ls -lh ubuntu.img
-rw-r--r-- 1 root root 10G Mar 11 11:54 ubuntu.img
```

The next step is to install Ubuntu (I already have a downloaded Ubuntu 9.10 (Karmic) ISO image in my current working directory):

```
$ qemu -hda ubuntu.img -boot c -cdrom ./ubuntu-9.10-desktop-i386.iso -m 512
```

In the above command, the *-hda* option specifies the disk image file; and *-cdrom* is the CD-ROM or ISO image to use as the 'optical drive' for the VM. The *-m* option specifies how much RAM this virtual machine will be given to use (in this case, I have allocated 512 MB of RAM; your choice should be based on your needs and hardware). Finally, we instruct QEMU to boot the VM from the ISO image by using the *-boot d* option. Once you run the above command, the VM will boot up and present the Ubuntu boot menu (see Figure 2). Follow the same installation steps you would use on a real machine. Once installed, you can boot the VM from the disk image with the following code: